MOVEINSYNC ENGINEERING ACTION PLAN & SPRINT SPECS

Product specifications, acceptance criteria, and 14-day sprint roadmap

MIS-001: SOS REACHABLE IN ONE TAP, NO BLOCKING POP-UPS ON THE TRIP SCREEN (SAFETY ESCALATION)

Component / Owner:	app_technical | Android lead
Effort / Priority:	Small Effort | Priority: Highest
Affected Flow & Versions:	Active Trip screen and Emergency SOS trigger (Versions: Unversioned / general (e.g. review b1814d48-3e54-4903-a109-e09e05bc3145))
Review Evidence:	*[b1814d48-3e54-4903-a109-e09e05bc3145] 'If someone needs to use SOS button, he/she can not use in timely manner due to useless notification pop ups'*
Hypotheses:	• App dialog manager renders non-dismissible modal notifications over the active trip view, overlaying and blocking the touch target of the SOS floating action button. • Notification stack fails to respect emergency view hierarchy z-indexing during ride state transitions.
Proposed Change:	Enforce top-level z-index window for SOS button that bypasses any in-app notification overlays. Implement persistent 1-tap SOS in system notification bar during active rides.
Acceptance Criteria:	1. SOS button is tap-responsive within 500ms under any active modal or notification state. 2. In-app popups cannot obscure the emergency action container. 3. Verified in automated Espresso UI tests.
Instrumentation Events:	trip_sos_tap, trip_sos_rendered, notification_popup_displayed
Internal Data Needed:	Client-side SOS latency logs and modal dismissal telemetry on active trip screen.
First 14 Days Plan:	Days 1–3: Inspect WindowManager and DialogFragment hierarchy on trip activity. Days 4–8: Refactor SOS to persistent non-blocking overlay. Days 9–14: Test on multiple Android OEM overlays and ship hotfix.
Metrics to Measure:	Zero blocked SOS events; 100% successful trigger within 1 tap under stress testing.

MIS-002: FIX APP NOT OPENING OR GETTING STUCK AFTER UPDATES; REPRODUCE UPGRADE PATH; STAGED ROLLOUT GATE

Component / Owner:	app_technical | Android lead + QA
Effort / Priority:	Medium Effort | Priority: High
Affected Flow & Versions:	App Cold Launch & Post-Upgrade SQLite/SharedPrefs Migration (Versions: 12.19.0 (10 low-star), 12.17.0 (8 low-star), 12.18.1 (5 low-star), 12.14.0 (3.37★ average))
Review Evidence:	*[60798e45-939e-4a99-8f5b-17383a891db6] 'whenever i update this app, i am not able to open the app again, I need to uninstall and reinstall again.' [7781ccc3-e390-49cb-b574-fb3213c2e74a] 'application not working with pixel 8 android version 17... installed and uninstalled...still not working...open and close automatically' [e735e280-9992-470e-b8df-d9b4a0e1e214] 'Unable to open the app on my phone. closes as soon as I try to open'*
Hypotheses:	• Room/SQLite schema migration failure on upgrade causes unhandled SQLiteConstraintException or IllegalStateException during Application.onCreate(). • Cached authentication tokens or Room database corrupted on delta APK upgrade, forcing users to uninstall and reinstall to wipe app data. • Android 14/15/16 background initialization changes causing ANR on modern Pixel/OnePlus devices.
Proposed Change:	Wrap local database upgrade migrations with automated fallback recovery (repair cache rather than crash). Set up CI upgrade tests (v12.14 -> v12.18 -> v12.19). Enforce staged rollout gates (freeze if crash-free sessions drop below 99.5%).
Acceptance Criteria:	1. 100% of schema migration test matrices pass in automated CI across 5 prior versions. 2. Crash-free user rate >= 99.5% on canary 5% rollout. 3. Automatic database corruption recovery clears bad cache without crashing.
Instrumentation Events:	app_launch_start, db_migration_success, db_migration_failure, app_cold_start_completed
Internal Data Needed:	Crashlytics / Firebase stack traces grouped by app_version and Android OS for launch crashes.
First 14 Days Plan:	Days 1–4: Reproduce upgrade path in QA emulator matrix; analyze Crashlytics launch exceptions. Days 5–9: Fix Room schema migration and corrupted state handler. Days 10–14: Staged rollout (5% -> 20% -> 100%) with automated ANR/crash halt.
Metrics to Measure:	Crash-free user rate (target: >= 99.5%); reduction in update-related review complaints.

MIS-003: FIX LOGOUT LOOP AND OTP DELIVERY; ADD RESEND AND A 'YOUR COMPANY HAS NOT ADDED YOU' MESSAGE

Component / Owner:	app_technical | Backend/auth lead
Effort / Priority:	Small Effort | Priority: High
Affected Flow & Versions:	Authentication, Session Renewal, and Employee Account Onboarding (Versions: 12.17.0 (6 low-star), 12.19.0 (2 low-star), 11.5.21 (1 low-star))
Review Evidence:	*[72e74b90-1da1-49cc-87c0-6bc42af65ba3] 'Pathetic . it logs out every time.'* *[c9c2d1f3-da5a-457f-b3cd-479afddacd8a] 'I\'m not able to login as it was automatically logged out and no otp received with registered mobile number'* *[d6cbf0b5-c7e3-4474-8a61-6a65d98ba8cf] 'Unable to login in the App. Both Mobile Otp and Email OTP options are not working'* *[d0013885-22ce-4c15-b188-eb1b58d912d3] 'doesn\'t send OTP, email, txt, WhatsApp no otps.'*
Hypotheses:	• Session refresh token expires prematurely or encounters 401 on background sync, causing silent logout back to welcome screen. • SMS gateway carrier routing failures or DND throttling drop OTP delivery without client fallback. • Unregistered enterprise employees attempt login before their corporate HR roster has imported their phone number, resulting in generic 'OTP failed' instead of clear explanation.
Proposed Change:	Graceful session refresh with silent token renewal. Add multi-channel OTP resend (SMS, WhatsApp, Email). If user number is not in enterprise roster, display explicit guidance: 'Your mobile number is not yet registered by your company HR transport team'.
Acceptance Criteria:	1. Multi-channel OTP fallback button appears within 30s. 2. Clear error banner for unrostered numbers explaining HR dependency. 3. Session token auto-renews without logging user out during active shifts.
Instrumentation Events:	auth_otp_requested, auth_otp_delivered, auth_otp_failed, auth_session_expired, auth_unregistered_employee
Internal Data Needed:	SMS vendor delivery receipt logs, telecom failure codes, enterprise HR roster sync audit tables.
First 14 Days Plan:	Days 1–3: Inspect auth service token TTL and 401 interception logic. Days 4–8: Implement WhatsApp/Email OTP fallback and explicit unrostered error messaging. Days 9–14: QA test and deploy to production auth service.
Metrics to Measure:	First-attempt login success rate (proposed target; baseline unknown); drop in OTP-related 1-star reviews.

LOW-HANGING REPUTATION & GROWTH LEVERS

Google In-App Review API Integration

Current State: Reviews are largely unprompted or prompted at friction points (e.g. ambiguous review complaint: 'forcing me to give rating after every ride'—needs verification whether in-app trip rating or Play prompt).

Recommendation: Trigger Google Play In-App Review API strictly after a high-satisfaction milestone: e.g., after 3 consecutive on-time trips completed without delay or dispute. Never prompt immediately after a delayed cab, cancelled booking, or app crash. Include an accessible in-app 'Report a Problem' button directly routing to the internal transport desk to intercept complaints before they reach the Play Store. Nothing is posted or changed automatically.

Human-in-the-Loop Developer Review Response SLA

Current State: Current reply rate on 1-2 star reviews is only 0.7%. Users feel abandoned.

Recommendation: Establish a dedicated 24-hour response SLA for all 1-2 star reviews. Every reply must be approved by a human operations lead (never automated canned bot responses). Acknowledge specific pain (e.g., 'We apologize for the delayed cab on your night shift'), provide a direct transport ticket ID, and invite them to update their rating once resolved. Nothing is posted automatically.

Staged Rollout & Crash-Rate Gating

Current State: Version analysis revealed sharp rating drops on specific versions (e.g., v12.14.0 lowest observed at 3.37★).

Recommendation: Adopt strict phased rollout gates: 5% day 1, 15% day 3, 50% day 5, 100% day 7. Freeze rollout immediately if crash-free user rate drops below 99.5% or Play Console ANR rate exceeds 0.47%.